Chapter 12, Problem 29P

Problem

A balanced three-phase Y-Δ system has $\mathbf{V}_{an} = 240\underline{/0^\circ}$ V rms and **Z**Δ = 51 + *j*45 Ω. If the line impedance per phase is 0.4 + *j*1.2 Ω, find the total complex power delivered to the load.

Step-by-step solution

Step 1 of 4

Consider the following balanced $Y - \Delta$ connection:

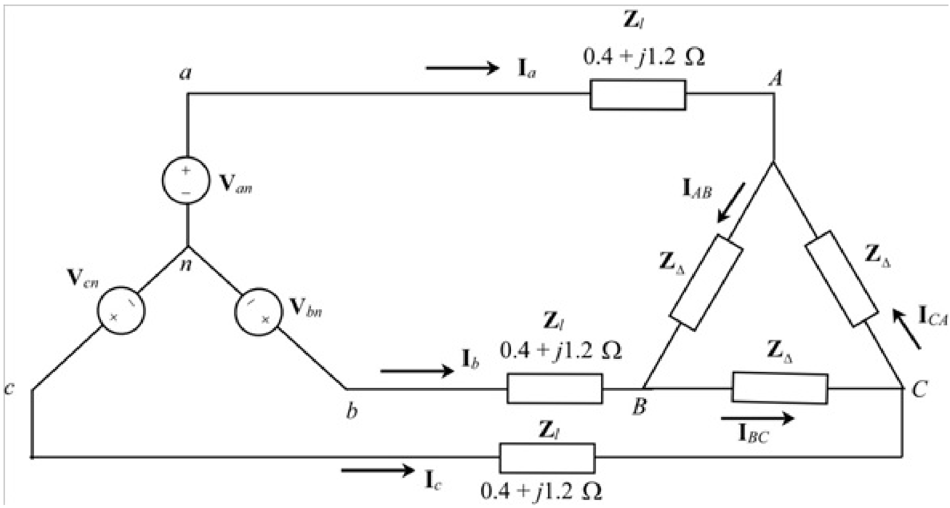

Figure 1

The load is delta connected. Convert delta load into equivalent Y-connected load.

The load impedance is,

$$\mathbf{Z}_\Delta = 51 + j45\,\Omega$$

Calculate the value of wye load.

$$\begin{aligned}\mathbf{Z}_Y &= \frac{\mathbf{Z}_\Delta}{3} \\ &= \frac{51 + j45}{3} \\ &= 17 + j15\,\Omega\end{aligned}$$

The modified circuit is shown in Figure 2.

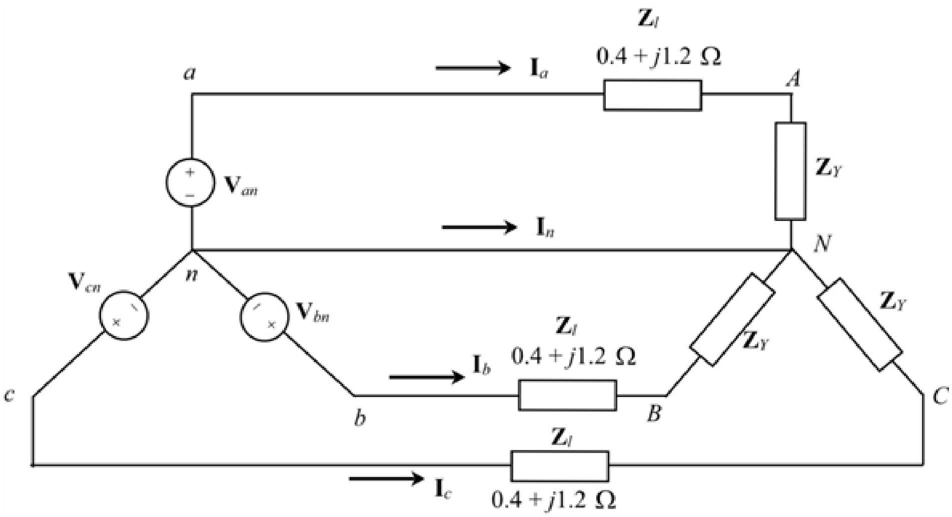

Figure 2

Step 2 of 4

Draw the per phase equivalent circuit as shown in Figure 3.

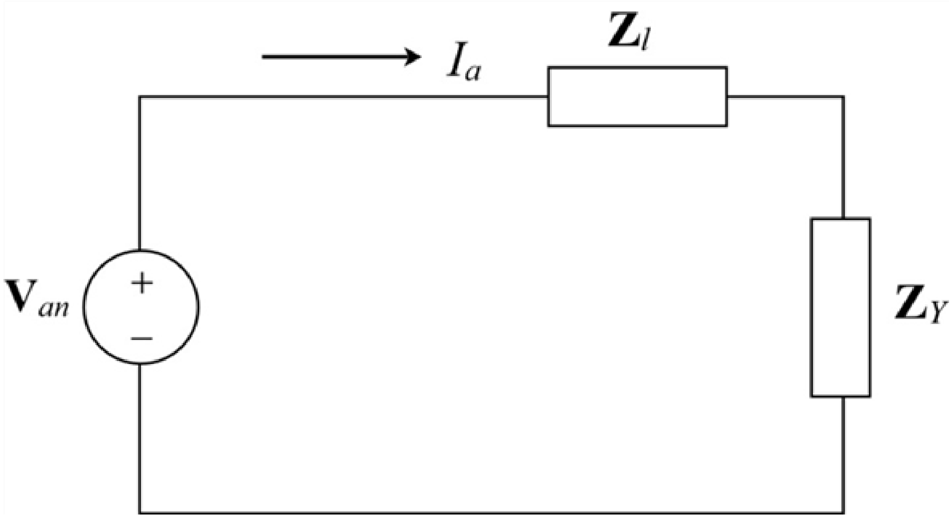

Figure 3

Step 3 of 4

Calculate the line current $\mathbf{I}_a$ from Figure 3.

$$\mathbf{I}_a = \frac{\mathbf{V}_{an}}{\mathbf{Z}_l + \mathbf{Z}_Y}$$

Substitute $240\angle 0°$ for $\mathbf{V}_{an}$, $0.4 + j1.2\,\Omega$ for $\mathbf{Z}_l$, and $17 + j15\,\Omega$ for $\mathbf{Z}_Y$.

$$\begin{aligned}\mathbf{I}_a &= \frac{240\angle 0°}{0.4 + j1.2 + 17 + j15} \\ &= \frac{240\angle 0°}{17.4 + j16.2} \\ &= \frac{240\angle 0°}{23.77\angle 42.95°} \\ &= 10.095\angle -42.95°\ \text{A}\end{aligned}$$

Calculate total complex power per phase delivered to the load, S.

$$\begin{aligned}S_{ph} &= \left|\mathbf{I}_a\right|^2 \mathbf{Z}_Y \\ &= (10.095)^2 (17 + j15) \\ &= 1732.45 + j1528.63\end{aligned}$$

Step 4 of 4

Calculate the total three phase complex power delivered to the load, S.

$$\begin{aligned}S &= 3S_{ph} \\ &= 3(1732.45 + j1528.63) \\ &= 5197.36 + j4585.91\ \text{VA} \\ &= 5.197 + j4.586\ \text{kVA}\end{aligned}$$

Therefore, the total complex power delivered to the load is $\boxed{5.197 + j4.586\ \text{kVA}}$.